

Healing Horizons: Adaptive VR for TBI Rehabilitation

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ABSTRACT

Healing Horizons presents a novel, bio-adaptive Virtual Reality (VR) approach to Traumatic Brain Injury (TBI) rehabilitation. Traditional therapies often fall short, prompting the need for a more personalised, dynamic intervention. Healing Horizons meets this demand by combining real-time physiological data with VR technology, facilitating effective recovery. The system uses the novel Galea VR head-mounted display (HMD) with integrated physiological sensors to continuously monitor brain activity, skin responses, muscle movements, and heart rate variability. These real-time bio-signals predict the user's cognitive fatigue, a common challenge in TBI patients. The VR environment then adapts dynamically to these physiological cues, ensuring the therapeutic experience is optimised for the individual's current cognitive state. This personalised approach helps develop resilience and social skills while addressing cognitive fatigue within a realistic yet controlled setting. Our demo showcases the potential of VR HMDs with physiological sensors to be used for bio-responsive virtual rehabilitation.

KEYWORDS

Virtual Reality, Biosignals, Traumatic Brain Injury, VR Therapy, Cognitive Fatigue

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1 INTRODUCTION

“Healing Horizons” (HH) presents an innovative Virtual Reality (VR) experience meticulously designed for the rehabilitation of individuals grappling with the consequences of Traumatic Brain Injury (TBI). Cognitive fatigue (CF), defined as the subjective lack of mental energy that incurs mental exhaustion from prolonged periods of cognitive activity [DeLuca 2005], is an unfortunate side effect of TBI. It renders everyday activities, such as socializing in a coffee shop, excessively exhausting for those affected.

Individuals with TBI must comprehend and adapt to their impairments to facilitate recovery. Consequently, TBI treatment frequently involves intensive interaction with clinicians, sometimes extending to visits to social spaces like cafes to practice coping mechanisms for CF. However, this approach can be time-consuming, expensive, and challenging to manage due to uncontrollable environmental stressors in public spaces, such as noise and crowding.

VR introduces a solution by providing a controlled setting for CF treatment. VR rehabilitation environments can be streamlined to ensure success compared to ‘real world’ training. For example, the stimuli presented in VR can be tightly controlled to reduce the stress and cognitive load on the user. VR interventions have been proven to enhance memory and cognition in TBI patients, with patients demonstrating greater motivation in VR settings compared to traditional rehabilitation contexts [Shin and Kim 2015].

A fundamental limitation of existing VR TBI therapy systems is the inability to adapt to the patient's CF during the VR experience. Current models uniformly train all patients, relying on self-reported cognitive load and stress levels. “HH” extends our previous work

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Figure 1: VR Experience: Welcoming the Patient



Figure 2: Picture of the Cafe VR experience

in the realm of VR for TBI rehabilitation [Nunnerley et al. 2022] by introducing a dynamic adaptation mechanism based on the user’s mental state, as monitored by biosensors in the VR head-mounted display (HMD). This feature positions “HH” as a pioneering tool in the landscape of a patient-centred approach to rehabilitation by using real-time physiological data to personalize and adapt the VR environment to the patient’s current cognitive state, effectively mitigating cognitive fatigue and stress.

Our ultimate objective is to empower clinicians to enhance the coping mechanisms of TBI patients against real-world stressful scenarios, providing them with a safe environment for practising effective CF management strategies. This system furnishes a secure, manageable setting conducive to acquiring crucial skills such as short-term and long-term memory, selective attention, management of distraction and fatigue, social interaction, information processing, and problem-solving.

2 DESIGN AND USER EXPERIENCE

In the “Healing Horizon” (HH) VR simulation, users engage in a bustling café scenario, interacting with virtual characters and performing tasks like ordering food to evaluate and address cognitive fatigue—a primary challenge for TBI patients. Users utilize the Galea VR Head-Mounted Display (HMD) from OpenBCI [OpenBCI 2023] (refer to figure 3), streaming real-time EEG, EMG, PPG, and EDA data. This data informs a predictive model, categorizing cognitive fatigue into levels such as low or severe. The experience begins at the café entrance, interacting with a Non-Player Character (NPC) to set a baseline physiological response (see figure 1). Inside the café, users mimic real-world social challenges TBI patients face through tasks, inducing cognitive stress.

The predictive model continuously assesses these activities, determining real-time cognitive fatigue. The VR scenario then responsively adjusts, offering a tailored rehabilitation experience aiding the user’s recovery. Contextually relevant audio-visual distractions,



Figure 3: OpenBCI’s Galea HMD with integrated biosensors

like a crying baby, loud conversations, and a passing police car with sirens, are presented in the VR environment. These elements, informed by prior research [Nunnerley et al. 2022], present cognitive challenges to the user, testing their resilience and ability to manage cognitive fatigue. While the user is in VR, they can be shown a real-time visualisation of their cognitive load and stress level, helping them see the impact of distractions and tasks.

3 PROTOTYPE IMPLEMENTATION

“Healing Horizons” (HH) actively quantifies and adapts to cognitive fatigue in therapeutic Virtual Reality (VR) environments. Designed for diverse VR settings, HH allows clinicians to tailor patient challenges, ranging from peaceful interactions to noisy distractions. The system integrates the BioSignal Processor (BP) with the Galea VR HMD, collecting real-time EEG, EMG, EDA, and PPG data. This data streams to a Python signal processor using the LabStreaming Layer (LSL) protocol. After processing, the Cognitive Fatigue Recognition (CFR) system, trained on data from 15 participants with over 80% accuracy, classifies fatigue levels. The results, from ‘low’ to ‘severe’, inform the VR’s Adaptor module via LSL. Depending on the detected fatigue, the Adaptor adjusts audio-visual distractions, ensuring the appropriate cognitive challenge for users.

4 CONCLUSION

“Healing Horizons” is a prototype platform for treating Cognitive Fatigue in TBI patients that adapts to their cognitive fatigue levels and provides personalized treatment. The current version demonstrates this in the virtual cafe, but in the future, other virtual environments will be created simulating everyday tasks, such as shopping in a supermarket. We have just completed clinical trials with the earlier non-adaptive application [Nunnerley et al. 2022] and plan to conduct formal user studies and clinical trials with this prototype version.

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